Assignment

Bachelor of Computer Applications

Semester III

Database Management System

Topic: Basic SQL Queries and Aggregate Functions

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Basic SQL Queries and Aggregate Functions

1. Objective

To retrieve and analyze data from the Hospital Management System using SQL queries. This assignment demonstrates filtering, sorting, grouping, filtering grouped results, and aggregate functions.

2. Introduction

SQL queries help retrieve useful information from a database. The WHERE clause filters records, ORDER BY sorts records, GROUP BY groups records, and HAVING filters grouped results. Aggregate functions perform calculations on data.

3. Filtering Records Using WHERE

Query 1: Display female patients

SQL command:

```
SELECT *
FROM Patient
WHERE gender = 'Female';
```

Explanation: This query displays only the patient records where the gender is Female.

patient_id	name	dob	gender	phone_no	address	blood_group	emergency_contact
7002	Diya Nair	2003-08-20	Female	9876547002	Marathahalli, Bengaluru	B+	9876548002
7004	Meera Joseph	2004-02-18	Female	9876547004	Whitefield, Bengaluru	AB+	9876548004

Query 2: Display scheduled appointments

SQL command:

```
SELECT *
FROM Appointment
WHERE status = 'Scheduled';
```

Explanation: This query displays appointments whose status is Scheduled.

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status	reason
7001	7001	7001	2026-10-26	10:00:00	Scheduled	Heart checkup
7002	7002	7002	2026-10-26	11:30:00	Scheduled	Child health checkup
7004	7004	7001	2026-10-28	14:00:00	Scheduled	Cardiac consultation
9002	1	1	2026-10-25	10:00:00	Scheduled	General consultation

Query 3: Display patients within a patient ID range

SQL command:

```
SELECT COUNT(*) AS total_patients
FROM Patient
WHERE patient_id BETWEEN 7001 AND 7004;
```

Explanation: This query counts patients whose IDs fall between 7001 and 7004, including both boundary values.

total_patients
4

4. Sorting Records Using ORDER BY

Query 4: Sort doctors by consultation fee

SQL command:

```
SELECT *
FROM Doctor
ORDER BY consultation_fee ASC;
```

Explanation: This query displays doctors in ascending order of consultation fees, from the lowest to the highest.

doctor_id	name	specialization	department_id	phone_no	email	experience_years	consultation_fee
7003	Ananya Rao	General Medicine	1	98765437003	ananya.rao@example.com	4	400.00
1	Dr. Arun	General Medicine	1	9876543211	arun.test@example.com	5	500.00
7002	Rahul Kumar	Pediatrics	7002	98765437002	rahul.kumar@example.com	6	500.00
7001	Priya Sharma	Cardiology	7001	98765437001	priya.sharma@example.com	8	800.00

Query 5: Sort appointments by date

SQL command:

```
SELECT *
FROM Appointment
ORDER BY appointment_date DESC;
```

Explanation: This query displays appointments in descending order of appointment date, with the latest dates first.

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status	reason
7004	7004	7001	2026-10-28	14:00:00	Scheduled	Cardiac consultation
7003	7003	7003	2026-10-27	09:30:00	Completed	General checkup
7001	7001	7001	2026-10-26	10:00:00	Scheduled	Heart checkup
7002	7002	7002	2026-10-26	11:30:00	Scheduled	Child health checkup
9002	1	1	2026-10-25	10:00:00	Scheduled	General consultation

5. Grouping Records Using GROUP BY

Query 6: Count doctors in each department

SQL command:

```
SELECT department_id, COUNT(*) AS total_doctors
FROM Doctor
GROUP BY department_id;
```

Explanation: This query groups doctors by department ID and counts the doctors in each department.

department_id	total_doctors
1	2
7001	1
7002	1

Query 7: Count appointments by status

SQL command:

```
SELECT status, COUNT(*) AS total_appointments
FROM Appointment
GROUP BY status;
```

Explanation: This query groups appointments according to their status and counts the appointments in each group.

status	total_appointments
Scheduled	4
Completed	1

6. Filtering Grouped Results Using HAVING

Query 8: Find departments with at least two doctors

SQL command:

```
SELECT department_id, COUNT(*) AS total_doctors
FROM Doctor
GROUP BY department_id
HAVING COUNT(*) >= 2;
```

Explanation: This query groups doctors by department and displays only departments with two or more doctors.

department_id	total_doctors
1	2

Query 9: Find manufacturers whose average medicine price is above 3

SQL command:

```
SELECT manufacturer, AVG(unit_price) AS average_price
FROM Medicine
GROUP BY manufacturer
HAVING AVG(unit_price) > 3;
```

Explanation: This query groups medicines by manufacturer, calculates the average price for each manufacturer, and displays only groups with an average price greater than 3.

manufacturer	average_price
HealthCare Labs	5.000000
MediCare Pharma	3.500000

7. Aggregate Functions

Aggregate functions calculate values from multiple records.

Query 10: COUNT() — Count patients

SQL command:

```
SELECT COUNT(*) AS total_patients
FROM Patient
WHERE patient_id BETWEEN 7001 AND 7004;
```

Explanation: COUNT() returns the number of patient records satisfying the specified condition.

total_patients
4

Query 11: SUM() — Calculate the total bill amount

SQL command:

```
SELECT SUM(total_amount) AS total_bill_amount
FROM Bill;
```

Explanation: SUM() adds the total amounts of all records in the Bill table.

total_bill_amount
6850.00

Query 12: AVG() — Calculate the average bill amount

SQL command:

```
SELECT AVG(total_amount) AS average_bill_amount
FROM Bill;
```

Explanation: AVG() calculates the average of the bill amounts.

average_bill_amount
2283.333333

Query 13: MIN() — Find the minimum medicine price

SQL command:

```
SELECT MIN(unit_price) AS minimum_medicine_price
FROM Medicine;
```

Explanation: MIN() returns the lowest medicine price recorded in the Medicine table.

minimum_medicine_price
2.50

Query 14: MAX() — Find the maximum medicine price

SQL command:

```
SELECT MAX(unit_price) AS maximum_medicine_price
FROM Medicine;
```

Explanation: MAX() returns the highest medicine price recorded in the Medicine table.

maximum_medicine_price
5.00

8. Conclusion

This assignment demonstrates how SQL can retrieve and analyze data in the Hospital Management System. The WHERE, ORDER BY, GROUP BY, and HAVING clauses help filter, sort, group, and analyze records. The aggregate functions COUNT(), SUM(), AVG(), MIN(), and MAX() provide useful numerical summaries of patient, doctor, appointment, medicine, and billing data.